Power: batteries and the sun

IoT · Level 3 · Lesson 3



Today we will

- Estimate battery life from capacity and average current.
- Explain why sleeping and sending less often make a sensor last longer.
- Size a small solar panel and battery for a sensor that runs all year.

Power for sensors

- Hours = capacity (mAh) ÷ average current (mA)
- Plan with 70 % of the label
- Sending costs most: send less often
- Solar: battery for cloudy days, panel 3 × daily use

Battery-life sums

1. Warm-up (5 min): The garden sensor that died in the holiday.
2. Teach (10 min): Capacity ÷ average current, with the 70 % rule.
3. Sums (12 min): Groups work out their sensor card's battery life.
4. Challenge (8 min): Change one thing to last a whole term.
5. Wrap-up (5 min): Size a battery and panel for cloudy days.

Prepare one sensor card per group and work the answers beforehand; a coin cell with Wi-Fi or LoRa makes a good trap card. Use batteries only, never mains. Never cut, crush or puncture lithium batteries; an adult can open an old solar garden light with a screwdriver to show the panel, battery and LED.

Let’s discuss

1. Which school sensor needs fresh data every minute, and which could report once an hour?
2. Where at school would a solar panel get full sun all year?
3. Who will clean the panel and check the battery each term?

Exit ticket

- A sensor averages 2 mA on 2,500 mAh of batteries. Planning with 70 %, about how many days will it last?